

VANESSA NGENO

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PROFESSIONAL SUMMARY

Data Analytics and Business Intelligence professional with 3.5 years of direct analytics and BI experience, complemented by 5 years in customer-facing operational roles where data analysis, workflow optimization, and process documentation were central to daily responsibilities.

Skilled in translating complex business needs into SQL and Python-based analytical solutions, building production-grade dashboards, and maintaining data governance standards across cross-functional environments.

Consistently delivers insights that drive strategic decisions, operational efficiency, and measurable business growth.

PROFESSIONAL EXPERIENCE

Data Analytics Intern

September 2025 – Present

Peblink, Remote

- Translated business requirements into analytical solutions including a 15-table relational data schema and 4,563-row synthetic dataset supporting a full correctional literacy program lifecycle
- Built a 5-layer interactive Tableau Public dashboard covering literacy progression, cognitive pattern detection, GED readiness, workforce completion, and post-release outcomes for DOC partner and funder reporting
- Designed proprietary composite KPIs including Overall Literacy Score, GED Readiness Score, and Program Effectiveness Score, establishing a licensable data asset with formal documentation for governance and transparency
- Applied machine learning models and GIS-based visualizations to identify literacy patterns and disparities across five countries, integrating datasets from UNESCO and World Bank with validation frameworks

Data Analytics Partner

September 2023 – February 2024

Safaricom PLC, Nairobi, Kenya

- Conducted customer behavior analysis using SQL, Python, and Qlik Sense, contributing to a 25% increase in retention and \$5M in cost savings
- Automated ETL reporting pipelines, reducing analysis turnaround time by 40% and improving decision-making efficiency
- Delivered actionable insights through dashboards and reports to support strategic business operations

Regional Business Intelligence Lead

January 2023 – August 2023

Safaricom PLC, Nairobi, Kenya

- Owned end-to-end analytics delivery across multiple business units, translating business needs into SQL and Python-based analytical solutions that drove a 30% increase in mobile money transactions and 12% growth in active subscribers
- Built and maintained Tableau and Power BI dashboards tracking KPIs for operational decision-makers, supporting a 10% reduction in customer complaints
- Developed and optimized SQL queries to produce performance-ready datasets for forecasting and reporting, enabling faster insight delivery across the organization
- Applied descriptive and inferential statistical methods to identify trends and anomalies, validated KPI definitions with stakeholders, and documented data flows to support governance and data quality standards
- Contributed to agile analytics delivery cycles, defining requirements iteratively with cross-functional teams across Finance, Sales, and Operations

Customer Experience Executive

October 2016 – December 2022

Safaricom PLC, Nairobi, Kenya

- Analyzed operational workflows and customer data to identify service bottlenecks and inefficiencies, collaborating across teams to implement process improvements
- Maintained structured documentation of processes and decision flows to ensure operational consistency and data integrity

ACADEMIC PROJECTS

DOC Literacy Dashboard Ecosystem — World Literacy Research Center - [Link](#) Spring 2026

- Designed relational schema, generated synthetic datasets using Python, and built a 5-layer Tableau Public dashboard with formal documentation for government and funder audiences
- Developed composite KPIs and data governance documentation supporting a licensable analytics product

AI-Powered Learning Assistant - [Link](#) Fall 2025

- Built production AI platform using Python, FastAPI, LangChain, RAG architecture, ChromaDB, and Claude API with analytics dashboards tracking user engagement and system performance

Abraham Accords Literacy Dashboard – Mapping Literacy for Humanity - [Link](#) Fall 2025

- Integrated multi-source international datasets with validation frameworks and built interactive Streamlit dashboards with forecasting modules for policy and executive audiences

EDUCATION

Yeshiva University, Katz School of Science and Health New York, NY
Master of Science in Data Analytics and Visualization, GPA: 3.4 December 2025

Jomo Kenyatta University of Agriculture and Technology Nairobi, Kenya
Bachelor of Science in Actuarial Science, GPA: 3.0 November 2015

SKILLS AND CERTIFICATIONS

Programming Languages: Python, R, SQL, JavaScript
Data Visualization & BI: Excel, Tableau, Power BI, Qlik Sense, Streamlit
Statistical Methods: Descriptive and Inferential Statistics, Hypothesis Testing, Regression Modeling, Time Series Forecasting, A/B Testing, Quantitative Analytics
Cloud & Tools: AWS, Docker, MongoDB, Railway, Databricks, BigQuery
AI/ML: Machine Learning, NLP, RAG Architecture, LangChain, ChromaDB, OpenAI API, Claude API
Other: ETL Pipelines, Data Governance, Agile Delivery, API Development, Data Quality

LEADERSHIP AND ACTIVITIES

- **Co-founder, Math Tutoring and Literacy Initiative** - Designed and delivered student-centered math programs; organized literacy outreach including donation of educational and Christian-based books to underserved children
- **Volunteer, National Tree Planting Day** - Environmental sustainability aligned with corporate ESG commitments